

Segment Attractiveness and Homogeneity (SAH) Score

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The SAH score is a metric ranging from 0 to 1, designed to quantify how well points are homogeneously spread within a defined segment $[a, b]$. An SAH score of 1 represents an ideal distribution.

1 SAH Calculation

Given a 1D array of points $P = \{p_1, p_2, \dots, p_n\}$, and segment boundaries a, b (with $a < b$). The segment length is $L = b - a$.

1. **Segment Attractiveness Weight Function** $w(p_i)$: For each point p_i , its distance $d_S(p_i)$ from the segment $[a, b]$ is calculated:

$$d_S(p_i) = \begin{cases} a - p_i & \text{if } p_i < a \\ 0 & \text{if } a \leq p_i \leq b \\ p_i - b & \text{if } p_i > b \end{cases}$$

The weight $w(p_i)$ is then given by:

$$w(p_i) = \exp\left(\frac{-2 \cdot d_S(p_i)^2}{L^2}\right)$$

2. **Mean Attractiveness (MA)**: If $n = 0$, $MA = 0.0$. Otherwise:

$$MA = \frac{1}{n} \sum_{i=1}^n w(p_i)$$

3. **Weighted Homogeneity (H_W)**: $H_W = 1.0$ if $n \leq 1$. Otherwise, points are sorted P_{sorted} , and corresponding weights are W_{sorted} . Let $W_{sum} = \sum w_i$. If $W_{sum} \approx 0$, $H_W = 1.0$. Nearest neighbor distances $d_{nn,i}$ are calculated for sorted points.

- Weighted mean of nearest neighbor distances (μ_d):

$$\mu_d = \frac{\sum (d_{nn,i} \cdot w_{sorted,i})}{W_{sum}}$$

- Weighted variance of nearest neighbor distances (σ_d^2):

$$\sigma_d^2 = \frac{\sum ((d_{nn,i} - \mu_d)^2 \cdot w_{sorted,i})}{W_{sum}}$$

- Weighted coefficient of variation (CV_d): If $\mu_d \approx 0$, $CV_d = 0.0$. Otherwise:

$$CV_d = \frac{\sqrt{\sigma_d^2}}{\mu_d}$$

The homogeneity score H_W is:

$$H_W = \frac{1.0}{1.0 + CV_d}$$

4. **Final SAH Score**:

$$SAH = MA \times H_W$$